higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ROADS
NQF LEVEL 4

(12030534)

22 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 5 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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QUESTION 1: ROAD DRAINAGE

1.1 State THREE reasons why all roads around the country need proper drainage. (3)

1.2 List FIVE places where longitudinal side drains are used and commonly placed when constructing a road. (5)

1.3 List FOUR important requirements that need to be followed and implemented when choosing and packing the correct type of stones (rocks) into a gabion basket. (4)

1.4 State FIVE problems that may arise and may be the cause of an insufficient subsurface drainage system on a road. (5)

1.5 List THREE requirements that are necessary for a manhole cover when constructing a manhole for a road drainage system. (3)

1.6 Explain the excavation requirements that need to be followed when excavating trenches for slotted agricultural drains. (5)

1.7 Answer the following in terms of geofabric material being used in road construction.

1.7.1 What is geofabric material made from? (2)

1.7.2 State THREE reasons/purposes for using geofabric material in road drainage. (3)

[30]

QUESTION 2: ROAD PAVEMENT

2.1 Name FIVE aspects and characteristics that rigid and flexible pavement surface courses must provide to ensure a comfortable surface course for the motorist, which will sustain high traffic volumes and high traffic loads. (5)

2.2 To successfully build a concrete pavement, one needs to ensure that the concrete is of exceptional quality, especially in terms of the correct selection of materials and mix proportions of the concrete.

Decide and state what the requirements of the concrete mix will be to ensure that the concrete will meet the desired quality requirements. (3)

2.3 Define each of the following terms used in terms of placing and working with concrete.

2.3.1 Workability (2)

2.3.2 Consistency (2)

2.4 Discuss FIVE situations where concrete block paving will excel in terms of cost and performance when compared with other types of pavements. (5)

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2.5 Name FOUR different grades of bitumen binders used to bind the aggregates (stones) when constructing asphalt road surfaces. (4)

2.6 List FOUR tests that can be conducted/used to determine the performance and quality levels of bitumen. (4)

2.7 List THREE advantages when using the following types of asphalt as a road surfacing material.

2.7.1 Continuously graded asphalt (3)

2.7.2 Gap-graded asphalt (3)

2.7.3 Open-graded asphalt (3)

2.8 Name the different layers of materials that will be applied onto the following types of surfacing seals:

2.8.1 Double seal (3)

2.8.2 Single seal (3)

[40]

QUESTION 3: ROAD MAINTANANCE AND REHABILITATION

3.1 Road maintenance is essential to ease traffic flow without unnecessary interruption.

3.1.1 Define the concept *road maintenance*. (2)

3.1.2 Give THREE examples of rigid pavement maintenance. (3)

3.2 Indicate whether the following statements are TRUE or FALSE. Write only 'True' or 'False' next to the question number (3.2.1–3.2.5) in the ANSWER BOOK.

3.2.1 A buffer zone can divert or separate traffic to ensure the safety of workers when doing road maintenance.

3.2.2 Traffic cones are only used to indicate a road detour at night.

3.2.3 Work areas must be made visible and be barricaded from traffic.

3.2.4 A cross-over should allow traffic to flow smoothly from its normal roadway position into a temporary position.

3.2.5 A transition area is the area in which drivers are no longer required to obey temporary traffic signs.

(5 × 1) (5)

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3.3 Describe THREE types of road reserve maintenance work and/or activities that need to be done on the side of the road. (3)

3.4 Asphalt overlays are meant to produce a more durable road surface, which will also provide a better and smoother driving surface.

Explain FIVE steps that must be followed when applying an asphalt overlay to an existing road surface, which requires an upgrade. (5)

3.5 Differentiate between the following types of road maintenance:

3.5.1 *Road reconstruction* (2)

3.5.2 *Road rehabilitation* (2)

3.6 Identify the causes for the following types of failure, which will require full-depth repairs to the surface of a concrete road pavement:

3.6.1 Transverse cracking (2)

3.6.2 Shattered slabs and corner breaks (2)

3.6.3 Punch-outs (2)

3.6.4 Deep spalling (2)

[30]

TOTAL: 100

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